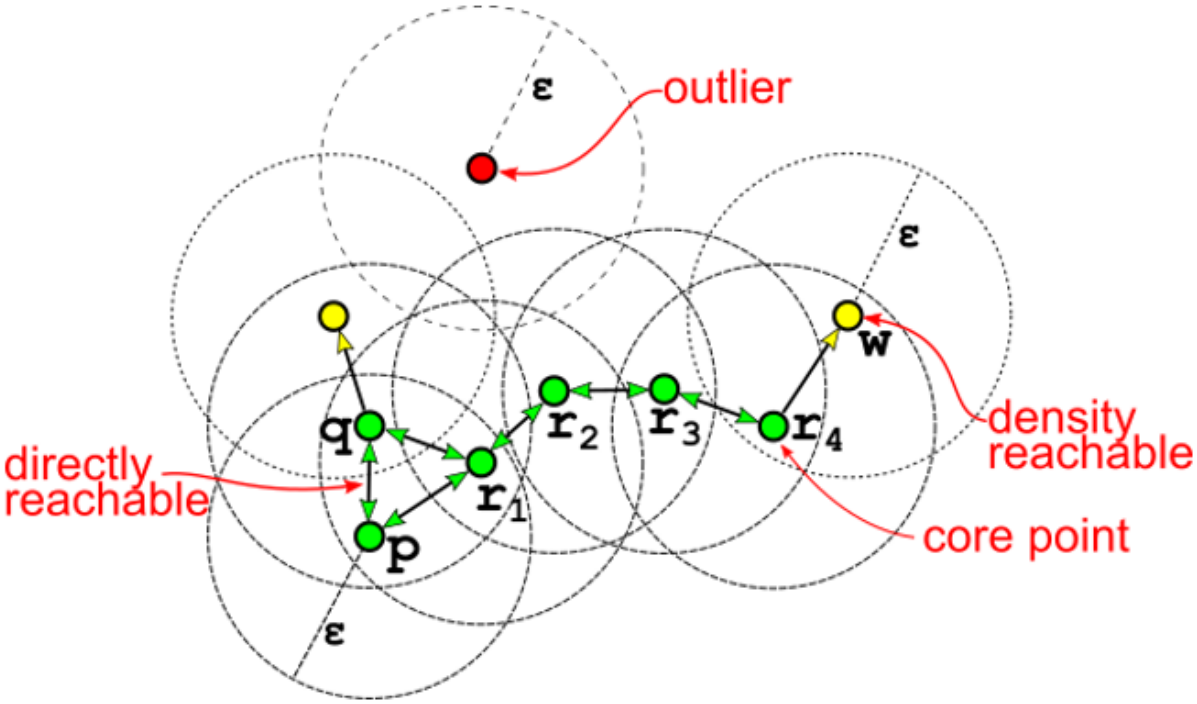


Figure 12.9 Core points, directly reachable points, and reachable points, given a radius ε and a threshold `minPoints` (the minimum number of points in a core region) equal to 3; hence, core points need to have at least 2 neighbors within distance ε.

The algorithm is built around the concept of **core points**: each of them has at least a certain number of neighbors within a specific distance; this can be seen from a different angle as well: core points are points in areas with at least a minimum density.

Core points (like p, q etc… in figure 12.9) that are reachable (meaning, adjacent) to each other belong to the same cluster. Why? Because we conjecture that high-density areas (as opposed to the low-density majority of the domain) define clusters. But all points that are within a distance ε from a core point p belongs to the same cluster as p's too.